

SET 13

1. R: Calculate mid-range and coefficient of range for 16,22,29,35,41,48,55.

R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
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Platform: x86_64-w64-mingw32/x64

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Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

```
> x <- c(16,22,29,35,41,48,55)
>
> # Mid-range
> mid_range <- (min(x) + max(x)) / 2
>
> # Coefficient of range
> coefficient_range <- (max(x) - min(x)) / (max(x) + min(x))
>
> print(mid_range)
[1] 35.5
> print(coefficient_range)
[1] 0.5492958
> |
```

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Values
coefficient_range 0.549295774647887
mid_range 35.5
x num [1:7] 16 22 29 35 41 48 55

2. R: Create five departments with employee counts and draw vertical and horizontal bar plots.

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[Workspace loaded from ~/.RData]

```
> departments <- c("HR", "IT", "Finance", "Sales", "Marketing")
> employees <- c(20, 35, 25, 40, 30)
>
> # Vertical bar plot
> barplot(employees,
+         names.arg = departments,
+         main = "Employees by Department",
+         xlab = "Departments",
+         ylab = "Employee Count")
>
> # Horizontal bar plot
> barplot(employees,
+         names.arg = departments,
+         main = "Employees by Department",
+         xlab = "Employee Count",
+         horiz = TRUE)
> |
```

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Values
departments chr [1:5] "HR" "IT" "Finance" "Sales" "Marketing"
employees num [1:5] 20 35 25 40 30

Employees by Department

Department	Employee Count
Marketing	30
Sales	40
Finance	25
IT	35
HR	20

3. WEKA: Create supermarket transactions; apply Apriori with confidence 80% using two minimum-support values and compare rule counts.

Dataset: Supermarket Transactions

```
@relation supermarket
@attribute Bread {T,F}
@attribute Milk {T,F}
@attribute Coffee {T,F}
@attribute Biscuits {T,F}
@attribute Soap {T,F}
@data
T,T,T,F,F
T,T,F,T,F
T,F,T,T,F
T,T,T,T,F
F,T,T,F,T
T,T,F,F,T
F,T,T,T,T
T,F,F,T,T
T,T,T,F,T
F,T,T,T,F
```

The screenshot shows the Weka Explorer interface with the 'Associate' tab selected. The 'Apriori' algorithm is chosen with the following command: `Apriori -N 10 -T 0 -C 0.8 -D 0.05 -U 0.2 -M 0.2 -S -1.0 -c -1`. The 'Associator output' pane displays the results of the algorithm, including the minimum support and confidence values, the number of cycles performed, the size of the set of large itemsets for each level, and the best rules found.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.8 -D 0.05 -U 0.2 -M 0.2 -S -1.0 -c -1

Start Stop

Result list (right-click for ...)

14:09:13 - Apriori

Associator output

=== Associator model (full training set) ===

Apriori

=====

Minimum support: 0.25 (3 instances)
Minimum metric <confidence>: 0.8
Number of cycles performed: 15

Generated sets of large itemsets:

Size of set of large itemsets L(1): 10
Size of set of large itemsets L(2): 28
Size of set of large itemsets L(3): 27
Size of set of large itemsets L(4): 9

Best rules found:

1. Milk=F 2 ==> Bread=T 2 <conf:(1)> lift:(1.43) lev:(0.06) [0] conv:(0.6)
2. Milk=F 2 ==> Biscuits=T 2 <conf:(1)> lift:(1.67) lev:(0.08) [0] conv:(0.8)
3. Milk=T Coffee=F 2 ==> Bread=T 2 <conf:(1)> lift:(1.43) lev:(0.06) [0] conv:(0.6)
4. Milk=F Biscuits=T 2 ==> Bread=T 2 <conf:(1)> lift:(1.43) lev:(0.06) [0] conv:(0.6)
5. Bread=T Milk=F 2 ==> Biscuits=T 2 <conf:(1)> lift:(1.67) lev:(0.08) [0] conv:(0.8)
6. Milk=F 2 ==> Bread=T Biscuits=T 2 <conf:(1)> lift:(2.5) lev:(0.12) [1] conv:(1.2)
7. Coffee=F Biscuits=T 2 ==> Bread=T 2 <conf:(1)> lift:(1.43) lev:(0.06) [0] conv:(0.6)
8. Coffee=F Soap=T 2 ==> Bread=T 2 <conf:(1)> lift:(1.43) lev:(0.06) [0] conv:(0.6)
9. Bread=F Biscuits=T 2 ==> Milk=T 2 <conf:(1)> lift:(1.25) lev:(0.04) [0] conv:(0.4)
10. Bread=F Soap=T 2 ==> Milk=T 2 <conf:(1)> lift:(1.25) lev:(0.04) [0] conv:(0.4)

Status OK Log x 0

4. WEKA: Apply FP-Growth to the supermarket data with two support settings and explain the effect on rule count.

Dataset: Supermarket Transactions – FP-Growth

Use the same Supermarket transaction ARFF dataset from Set 13 Q3.

The screenshot shows the Weka Explorer application window. The 'Associate' tab is selected. The 'Choose' dropdown is set to 'FPGrowth' with the command line: `-P 2 -I -1 -N 10 -T 0 -C 0.8 -D 0.05 -U 0.2 -M 0.2`. The 'Start' button is visible. The 'Result list' on the left shows two entries: '14:09:13 - Apriori' and '14:11:12 - FPGrowth', with the latter selected. The 'Associator output' pane displays the following text:

```
=== Run information ===  
  
Scheme:      weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.8 -D 0.05 -U 0.2 -M 0.2  
Relation:    supermarket  
Instances:   10  
Attributes:  5  
             Bread  
             Milk  
             Coffee  
             Biscuits  
             Soap  
  
=== Associator model (full training set) ===  
  
No rules found!
```

The status bar at the bottom shows 'Status OK' and a 'Log' button.

SET 14

1. R: Generate 30 normal observations with `set.seed(40)`; calculate mean and SD, draw histogram and density curve.
2. R: Create `EngineSize` and `FuelConsumption` vectors; draw scatter plot and calculate correlation.
3. WEKA: Create crop-classification ARFF data (`SoilType`, `Rainfall`, `Temperature`, `Crop`); apply J48 and visualize.

Dataset: Crop Classification

```
@relation crop
@attribute SoilType {Sandy,Loamy,Clay}
@attribute Rainfall numeric
@attribute Temperature numeric
@attribute Crop {Rice,Wheat,Millet}
@data
Clay,1200,28,Rice
Loamy,900,24,Wheat
Sandy,500,30,Millet
Clay,1100,27,Rice
Loamy,850,22,Wheat
Sandy,450,31,Millet
Clay,1300,29,Rice
Loamy,800,23,Wheat
Sandy,600,30,Millet
```

4. WEKA: Apply SMO to the crop data and compare precision, recall and F1 with J48.

Dataset: Crop Classification – SMO

Use the same Crop Classification ARFF dataset from Set 14 Q3.

SET 15

1. R: Calculate range, IQR, variance and SD for 10 transaction amounts and discuss sensitivity to an extreme value.
2. R: Create StudentID, Attendance, Marks and Grade data; use summary() and export CSV.
3. WEKA: Create telecom ARFF data (Age, MonthlyCharge, ContractType, Churn); apply J48 and report confusion matrix.

Dataset: Telecom Churn

```
@relation telecom_churn
@attribute Age numeric
@attribute MonthlyCharge numeric
@attribute ContractType {Monthly,Yearly}
@attribute Churn {Yes,No}
@data
22,45,Monthly,Yes
28,55,Monthly,Yes
35,60,Yearly,No
42,70,Yearly,No
25,50,Monthly,Yes
38,65,Yearly,No
31,58,Monthly,No
47,75,Yearly,No
```

4. WEKA: Apply Naive Bayes to telecom data and compare correct percentage and Kappa with J48.

Dataset: Telecom Churn – Naïve Bayes

Use the same Telecom Churn ARFF dataset from Set 15 Q3.

SET 16

1. R: Find Q1 and Q3 for 11,14,16,18,21,23,26,29,31,34,38,42,47 and draw a boxplot.
2. R: Demonstrate vector addition, subtraction, multiplication and division on two numeric vectors.
3. WEKA: Create vehicle-classification ARFF data (EnginePower, Weight, Mileage, VehicleType); apply J48 and inspect rules.

Dataset: Vehicle Classification

```
@relation vehicle
@attribute EnginePower numeric
@attribute Weight numeric
@attribute Mileage numeric
@attribute VehicleType {Hatchback,Sedan,SUV}
@data
70,900,20,Hatchback
90,1100,18,Sedan
120,1500,14,SUV
80,950,19,Hatchback
100,1250,16,Sedan
140,1800,12,SUV
75,920,21,Hatchback
110,1300,15,Sedan
150,1900,11,SUV
```

4. WEKA: Apply IBk to the vehicle data with two different K values and compare accuracy.

Dataset: Vehicle Classification – IBk

Use the same Vehicle Classification ARFF dataset from Set 16 Q3.

SET 17

1. R: Create a 4×4 marks matrix and calculate row totals, column totals and overall mean.
2. R: Calculate mean absolute deviation from the median for 9,12,15,18,20,24,28,31.
3. WEKA: Create credit-approval ARFF data (Income, Debt, EmploymentStatus, Approved); apply Naive Bayes and record evaluation.

Dataset: Credit Approval

```
@relation credit_approval
@attribute Income numeric
@attribute Debt numeric
@attribute EmploymentStatus {Employed,SelfEmployed,Unemployed}
@attribute Approved {Yes,No}
@data
25000,12000,Employed,No
35000,10000,Employed,Yes
45000,8000,Employed,Yes
30000,18000,SelfEmployed,No
55000,9000,SelfEmployed,Yes
22000,15000,Unemployed,No
48000,7000,Employed,Yes
38000,11000,SelfEmployed,Yes
```

4. WEKA: Apply SMO to the credit data and compare ROC area and F1 with Naive Bayes.

Dataset: Credit Approval – SMO

Use the same Credit Approval ARFF dataset from Set 17 Q3.

SET 18

1. R: Create monthly revenue for two years; draw a comparative line graph and find the largest year-to-year increase.
2. R: Calculate coefficient of variation for two datasets and identify greater relative variability.
3. WEKA: Create medical-screening ARFF data (BloodPressure, Cholesterol, Age, Risk); apply J48 and visualize.

Dataset: Medical Screening

```
@relation medical_screening
@attribute BloodPressure numeric
@attribute Cholesterol numeric
@attribute Age numeric
@attribute Risk {Low,High}
@data
118,170,25,Low
135,220,45,High
125,190,35,Low
145,240,55,High
130,210,42,High
115,165,28,Low
150,250,60,High
122,180,32,Low
```

4. WEKA: Apply Logistic classification to the medical data and compare accuracy and ROC area with J48.

Dataset: Medical Screening – Logistic

Use the same Medical Screening ARFF dataset from Set 18 Q3.

SET 19

1. R: Create 15 ages and divide them into three equal-frequency groups using quantile-based cut points.
2. R: Perform equal-width discretization of 10,14,18,23,29,34,41,48,55,63 into four intervals and show frequencies.
3. WEKA: Create retail ARFF data and use Discretize to convert numeric Income into five bins; apply Apriori.

Dataset: Retail Customer Discretization

```
@relation retail_customer
@attribute Age numeric
@attribute Income numeric
@attribute Visits numeric
@attribute Purchase {Yes,No}
@data
21,22000,4,No
25,28000,6,No
29,35000,8,Yes
33,42000,10,Yes
38,50000,12,Yes
45,62000,15,Yes
52,75000,16,Yes
27,30000,7,No
```

4. WEKA: Apply FP-Growth to the discretized retail data and compare rule count/confidence with Apriori.

Dataset: Retail Customer – FP-Growth

Use the discretized Retail Customer dataset from Set 19 Q3 after applying WEKA's Discretize filter.

SET 20

1. R: Create two classes of marks; compare means, medians and SDs, draw boxplots and identify the more consistent class.
2. R: Create Age and BodyFat for 12 adults; draw scatter plot, calculate Pearson correlation and create a Q-Q plot for Age.
3. WEKA: Create customer-purchase ARFF data (Age, Income, OnlineVisits, Purchased); compare J48 and SMO using 10-fold CV, accuracy and F1.

Dataset: Customer Purchase Comparison

```
@relation customer_purchase_compare
@attribute Age numeric
@attribute Income numeric
@attribute OnlineVisits numeric
@attribute Purchased {Yes,No}
@data
21,24000,3,No
26,32000,5,No
30,45000,8,Yes
35,52000,10,Yes
41,65000,13,Yes
48,78000,15,Yes
24,29000,4,No
38,58000,11,Yes
55,85000,16,Yes
32,47000,7,No
```

4. WEKA: Create at least 10 transactions; run Apriori and FP-Growth with identical support/confidence, then identify common and unique rules.

Dataset: Transaction Association Comparison

```
@relation market_transactions
@attribute Bread {T,F}
@attribute Milk {T,F}
@attribute Eggs {T,F}
@attribute Butter {T,F}
@attribute Juice {T,F}
@data
T,T,F,F,F
T,T,T,F,F
T,F,T,T,F
T,T,T,T,F
F,T,T,F,T
T,T,F,F,T
F,T,T,T,T
T,F,F,T,T
T,T,T,F,T
F,T,T,T,F
```